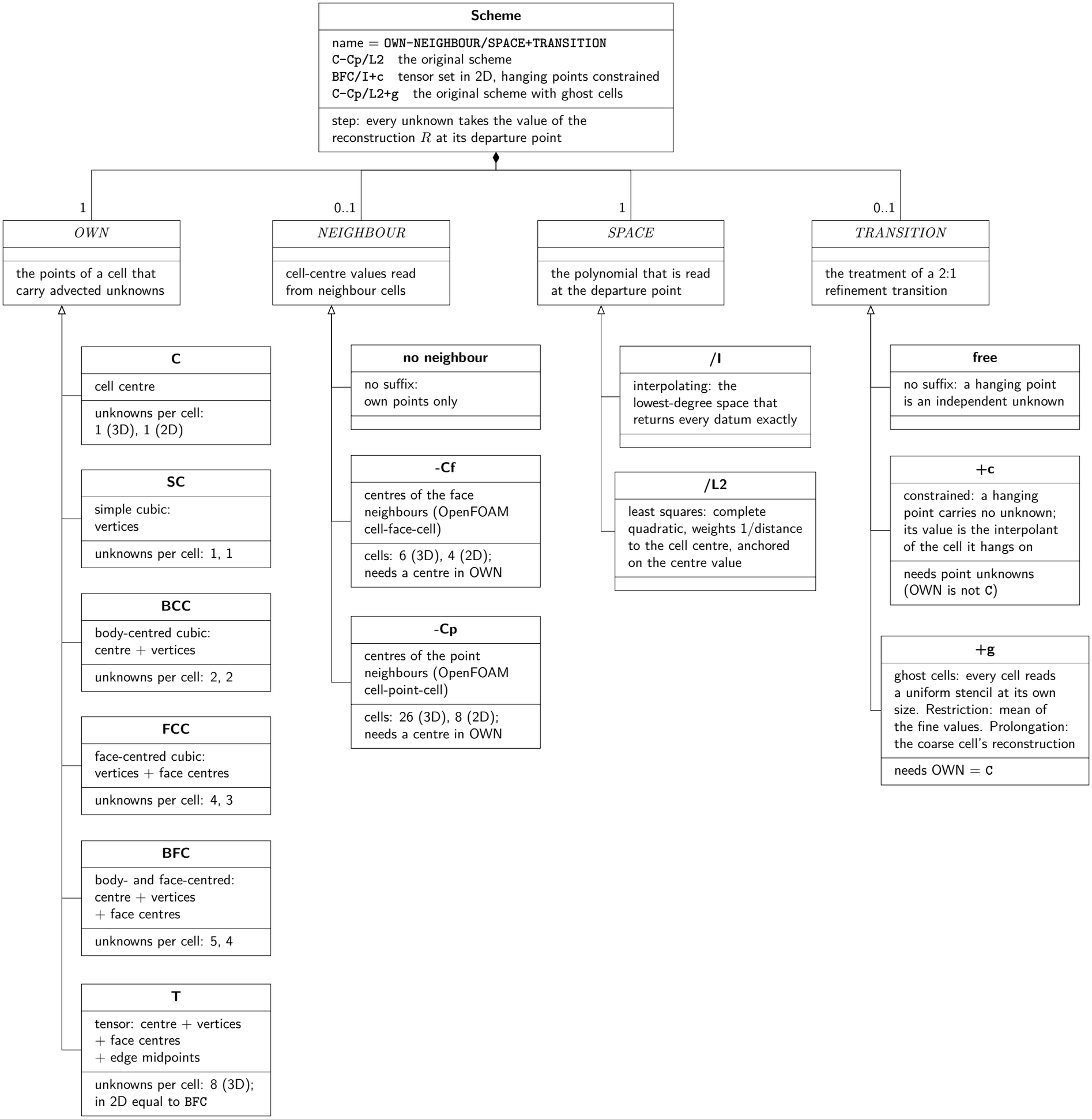

Scheme
name = OWN-NEIGHBOUR/SPACE+TRANSITION
C-Cp/L2 the original scheme
BFC/I+c tensor set in 2D, hanging points constrained
C-Cp/L2+g the original scheme with ghost cells
step: every unknown takes the value of the reconstruction R at its departure point
1
0..1
1
0..1
OWN
the points of a cell that carry advected unknowns
NEIGHBOUR
cell-centre values read from neighbour cells
SPACE
the polynomial that is read at the departure point
TRANSITION
the treatment of a 2:1 refinement transition
C
cell centre
unknowns per cell: 1 (3D), 1 (2D)
SC
simple cubic: vertices
unknowns per cell: 1, 1
BCC
body-centred cubic: centre + vertices
unknowns per cell: 2, 2
FCC
face-centred cubic: vertices + face centres
unknowns per cell: 4, 3
BFC
body- and face-centred: centre + vertices + face centres
unknowns per cell: 5, 4
T
tensor: centre + vertices + face centres + edge midpoints
unknowns per cell: 8 (3D); in 2D equal to BFC
no neighbour
no suffix: own points only
-Cf
centres of the face neighbours (OpenFOAM cell-face-cell)
cells: 6 (3D), 4 (2D); needs a centre in OWN
-Cp
centres of the point neighbours (OpenFOAM cell-point-cell)
cells: 26 (3D), 8 (2D); needs a centre in OWN
/I
interpolating: the lowest-degree space that returns every datum exactly
/L2
least squares: complete quadratic, weights 1/distance to the cell centre, anchored on the centre value
free
no suffix: a hanging point is an independent unknown
+c
constrained: a hanging point carries no unknown; its value is the interpolant of the cell it hangs on
needs point unknowns (OWN is not C)
+g
ghost cells: every cell reads a uniform stencil at its own size. Restriction: mean of the fine values. Prolongation: the coarse cell's reconstruction
needs OWN = C